



BITS Pilani
DIGITAL
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Module 2: Data Engineering

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Section 1: Opening and Framing

Data First AI System Scoping

Architecture is the consequence of data and constraints.

By the end of this session, you will be able to:

- Scope an AI problem through Who/What/Why/Where
- Map problem -> Model family
- Select reference architecture as a risk map
- Produce a defensible solution approach



Most AI Failures are Data Failures

- Discovered too late!
- Why do AI projects actually fail?
- Live Poll: What usually gets blamed?
 - ☐ Model
 - ☐ Data
 - ☐ Infrastructure
 - ☐ Business Clarity



Reframing AI System Design

Business Request $\neq$ AI Problem



AI Problem = Decision + Data + Risk

Section 2: Why Scoping Fails?

Common Failure Pattern

Teams start with:



“We want GenAI”

instead of

“What decisions are we supporting?”

Hidden Late Discoveries

Output quality undefined

Data assumed

Constraints discovered late

The Right Question



What decision - using what data - under what risk?

Section 3 - Data First Scoping Framework

The Four Questions



Who

What

Why

Where

Who

Stakeholders:

- End user
- Decision owner
- Impacted population
- Regulators

Drives:

- Fairness
- Explainability
- Audibility

What

System Function:

- Prediction
- Retrieval
- Ranking
- Generation
- Classification

Defines data shape

Why

Business Metric

ML Proxy Metric

Cost of Error

Defines:

- Evaluation
- Human Oversight

Where

Real time Vs batch

Scale

Latency

Regulatory Surface

Defines -> Pipeline and
governance

Key Synthesis

WHO -> Labeling & Fairness

WHY -> Error Tolerance

WHAT -> Schema

WHERE -> Pipeline &
Obervability

Section 4: Activity 1

Bad-Scoping Case: Retail Bank

A mid-sized retail bank is exploring a conversational AI assistant to support its loan approval teams as part of a broader digital transformation. The assistant would surface relevant applicant information, highlight key risk signals and provide a recommended credit decision with a natural-language explanation inside the existing loan processing system. The initiative aims to improve turnaround time, increase decision consistency and help onboard new credit analysts faster. The current proposal is to adapt a large language model using historical loan records, internal policy documents and customer interaction data, with near real-time responses that fit into today's workflow. The programme has executive sponsorship, access to enterprise data and a target rollout within two quarters, with the expectation that it will meet regulatory and model governance standards.

Breakout: Bad Scoping Detection

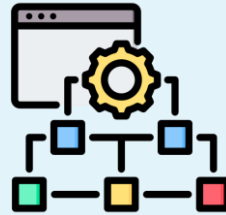
Your task:

Identify what is missing:

- Decision
- Data Realities
- Risks
- Constraint



Debriefing



Architecture does not come from tools, it emerges from constraints.

Section 5: Problem -> Model Family

We do not pick model!
We reason about model families.

Mapping Table

Problem shape -> Data -> Model Family

- Knowledge Lookup -> Document -> RAG
- Sequential Signals -> Time Series -> Forecasting
- Visual Inspection -> Images -> Vision Models
- Policy Decisions -> Mixed-Structured -> Hybrid ML
- Multi-Step Reasoning -> Tools Use -> Agents

Section 6: Reference Architecture as Risk Maps

Reference Architecture is NOT Diagrams

They show:

- Where data enters
- Where control lives
- Where system fails
- Where governance sits

When to Use What on What Risk

Batch ML:

Use when: Offline predictions

Risk: Stale data

Online Inference:

Use when: Real-time decisions

Risk: Latency and cascading failure

RAG:

Use when: Knowledge retrieval

Risk: Hallucination and stale content

Human-in-the-loop:

Use when: High-risk decisions

Risk: Automation bias

Agentic Workflow:

Use when: Multi-state reasoning

Risk: Unbounded cost

Section 7: Case Study

Case: Mid-Tier Hospital ED Triage System

Context:

A regional mid-tier hospital network (3 hospitals, ~600 beds total) is experiencing increasing emergency department (ED) congestion, delayed diagnoses and clinician burnout.

Leadership is exploring AI to improve patient triage and diagnostic support while maintaining regulatory compliance and patient safety.

Task: Prepare a data-first Architecture Decision Note (ADN)

The Questions and Decisions



- Which patients needs immediate attention?
- Who are the stake holders?
- What model family will be chosen?
- What constraints do you see?
- What is the best reference architecture?
- What are the trade offs you should consider?

Section 8: Activity 2 – Prepare an Architecture Decision Note

The Questions and Decisions

Each group must define:

- Decision
- Who/What/Why/Where
- Data
- Model family
- Architecture
- Risks
- Template-> Architecture Decision Note template

Debrief: Synthesis

Discussion points

Questions?

Trade-offs?

QUIZ

Take Home Assignment

Create a detailed Architecture Decision Note for the same case study or another case of your choice.



Thank You
